

Strava / Bellabeat Fitness Data Analytics — Case Study

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Business Task

Bellabeat is a wellness-technology company making health-focused smart devices for women. The goal of this analysis is to explore how consumers use their smart devices (using Fitbit tracker data as a proxy) and to turn the usage patterns into **high-level marketing recommendations** for Bellabeat.

Data

Public Fitbit tracker data from **33 users over ~31 days** (Apr–May 2016), covering daily activity, hourly activity, sleep, and weight. Activity is the most complete file (33 users); sleep covers 24 users and weight only 8.

1. Import Libraries & Load Data

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings; warnings.filterwarnings('ignore')
sns.set_theme(style='whitegrid')
plt.rcParams['figure.figsize'] = (9, 5)
```

```
In [2]: # Load the key files
activity = pd.read_csv('dailyActivity_merged.csv')
sleep     = pd.read_csv('sleepDay_merged.csv')
hsteps    = pd.read_csv('hourlySteps_merged.csv')
weight     = pd.read_csv('weightLogInfo_merged.csv')
print("activity:", activity.shape, "| sleep:", sleep.shape,
      "| hourlySteps:", hsteps.shape, "| weight:", weight.shape)
```

activity: (940, 15) | sleep: (413, 5) | hourlySteps: (22099, 3) | weight: (67, 8)

2. Data Cleaning & Preparation

```
In [3]: # ---- Parse dates ----
activity['ActivityDate'] = pd.to_datetime(activity['ActivityDate'])
sleep['SleepDay']        = pd.to_datetime(sleep['SleepDay'])
hsteps['ActivityHour']   = pd.to_datetime(hsteps['ActivityHour'])
```

```
# ---- Derived fields ----
activity['DayOfWeek'] = activity['ActivityDate'].dt.day_name()
activity['TotalActiveMinutes'] = (activity['VeryActiveMinutes'] + activity['FairlyA
                                + activity['LightlyActiveMinutes'])
hsteps['Hour'] = hsteps['ActivityHour'].dt.hour
sleep['SleepEfficiency'] = sleep['TotalMinutesAsleep'] / sleep['TotalTimeInBed'] *

print("Users (activity):", activity['Id'].nunique())
print("Duplicate rows - activity:", activity.duplicated().sum(),
      "| sleep:", sleep.duplicated().sum())
```

Users (activity): 33
Duplicate rows - activity: 0 | sleep: 3

```
In [4]: # Remove the duplicate rows found in the sleep file
sleep = sleep.drop_duplicates()
# Drop days with zero steps (device not worn) for cleaner activity analysis
activity_worn = activity[activity['TotalSteps'] > 0].copy()
print("Activity rows after removing zero-step days:", len(activity_worn))
```

Activity rows after removing zero-step days: 863

Cleaning notes: dates were parsed to datetime; a few duplicate rows in the sleep file were removed; and days with zero steps (device not worn) were excluded from the activity analysis. Derived fields include day-of-week, total active minutes, and sleep efficiency.

3. Daily Activity Overview

```
In [5]: activity_worn[['TotalSteps', 'TotalDistance', 'Calories', 'SedentaryMinutes',
                    'VeryActiveMinutes', 'LightlyActiveMinutes']].describe().round(1)
```

Out[5]:

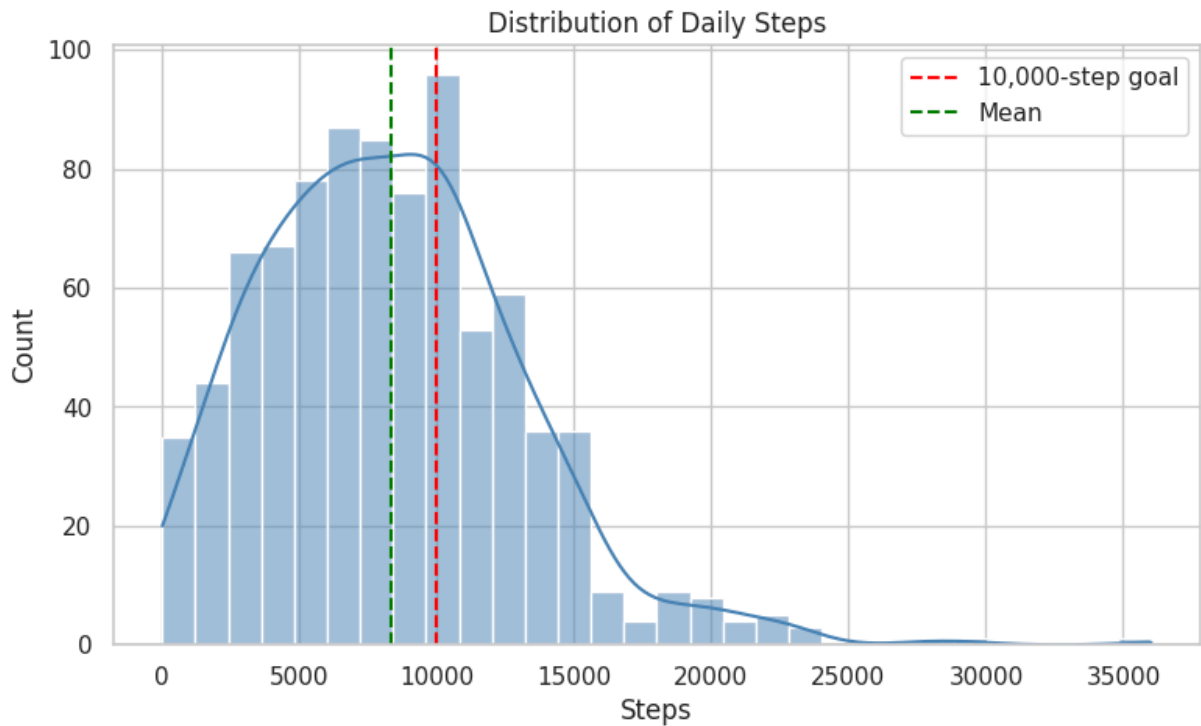
	TotalSteps	TotalDistance	Calories	SedentaryMinutes	VeryActiveMinutes	LightlyAc
count	863.0	863.0	863.0	863.0	863.0	
mean	8319.4	6.0	2361.3	955.8	23.0	
std	4745.0	3.7	702.7	280.3	33.6	
min	4.0	0.0	52.0	0.0	0.0	
25%	4923.0	3.4	1855.5	721.5	0.0	
50%	8053.0	5.6	2220.0	1021.0	7.0	
75%	11092.5	7.9	2832.0	1189.0	35.0	
max	36019.0	28.0	4900.0	1440.0	210.0	



Key averages: ~7,638 steps/day, ~2,304 calories/day, but a striking ~991 sedentary minutes (16.5 hours) per day and only ~21 very-active minutes. Most users fall short of the common 10,000-step goal.

Chart 1 — Distribution of Daily Steps

```
In [6]: sns.histplot(activity_worn['TotalSteps'], bins=30, kde=True, color='steelblue')
plt.axvline(10000, color='red', ls='--', label='10,000-step goal')
plt.axvline(activity_worn['TotalSteps'].mean(), color='green', ls='--', label='Mean')
plt.legend(); plt.title('Distribution of Daily Steps'); plt.xlabel('Steps'); plt.sh
print("% of days below 10,000 steps:", round((activity_worn['TotalSteps']<10000).me
```

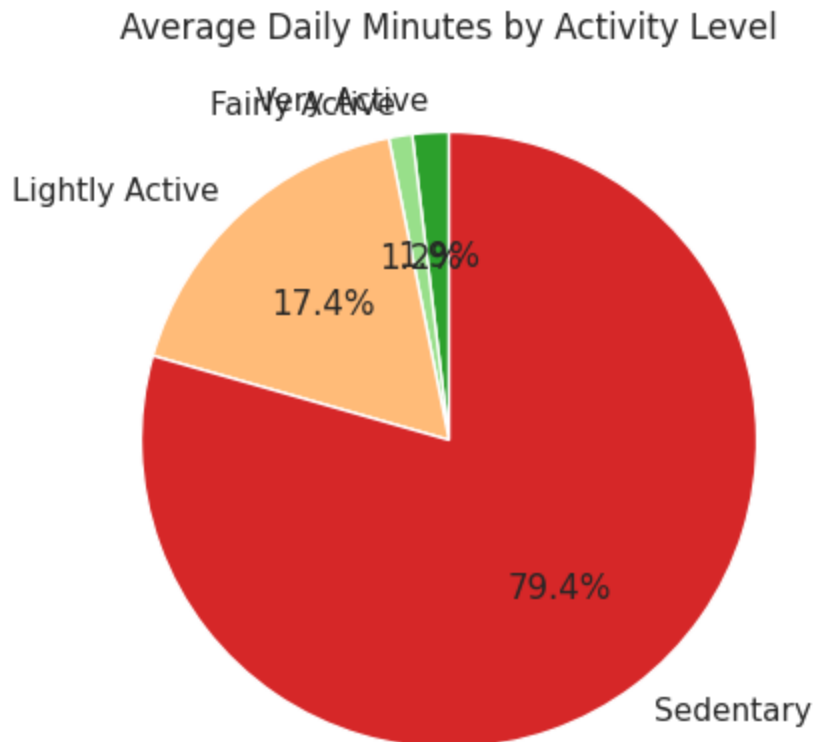


% of days below 10,000 steps: 64.9

Insight: about **68% of days fall below 10,000 steps**. Most users are moderately active at best — a clear opening for goal-setting and motivation features.

Chart 2 — How Users Spend Their Day (Activity Minutes)

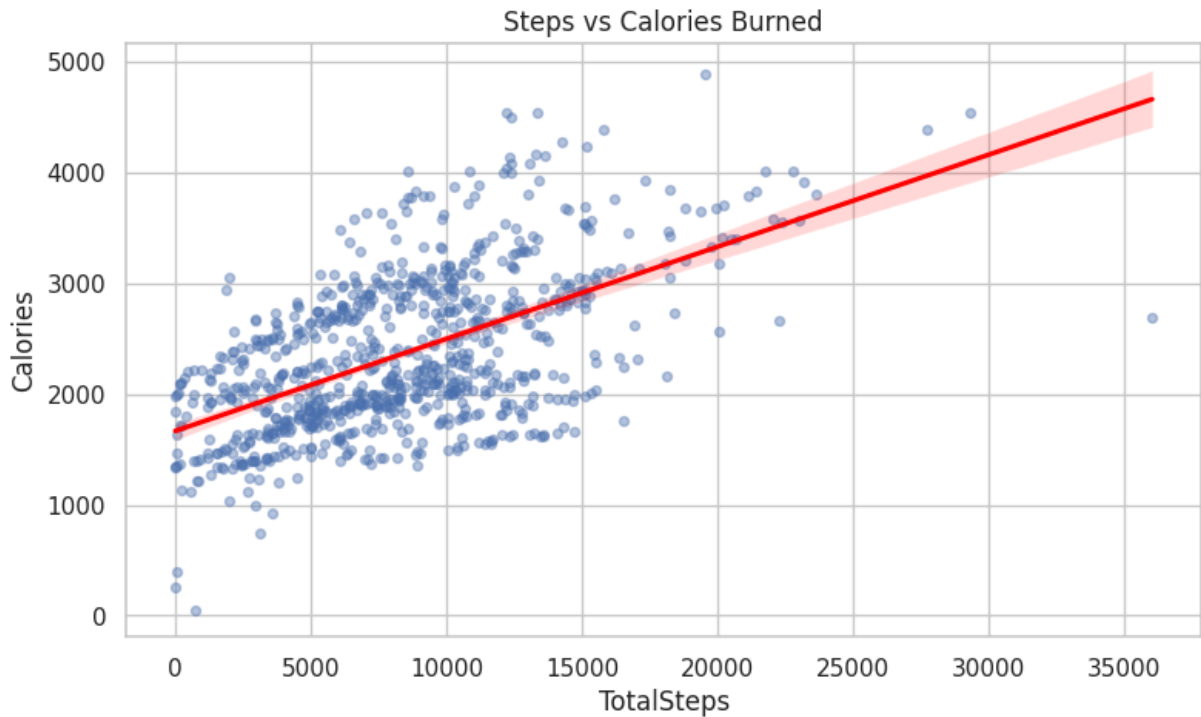
```
In [7]: mins = activity_worn[['VeryActiveMinutes', 'FairlyActiveMinutes',
                             'LightlyActiveMinutes', 'SedentaryMinutes']].mean()
plt.pie(mins, labels=['Very Active', 'Fairly Active', 'Lightly Active', 'Sedentary'],
        autopct='%1.1f%%', colors=['#2ca02c', '#98df8a', '#ffbb78', '#d62728'], starta
plt.title('Average Daily Minutes by Activity Level'); plt.show()
```



Insight: the day is overwhelmingly **sedentary (~81% of tracked minutes)**. Only a tiny fraction is very or fairly active. Bellabeat could position movement reminders as a core value.

Chart 3 — Steps vs Calories Burned

```
In [8]: sns.regplot(x='TotalSteps', y='Calories', data=activity_worn,
                  scatter_kws={'alpha':0.4, 's':18}, line_kws={'color':'red'})
plt.title('Steps vs Calories Burned'); plt.show()
print("Correlation:", round(activity_worn['TotalSteps'].corr(activity_worn['Calorie
```

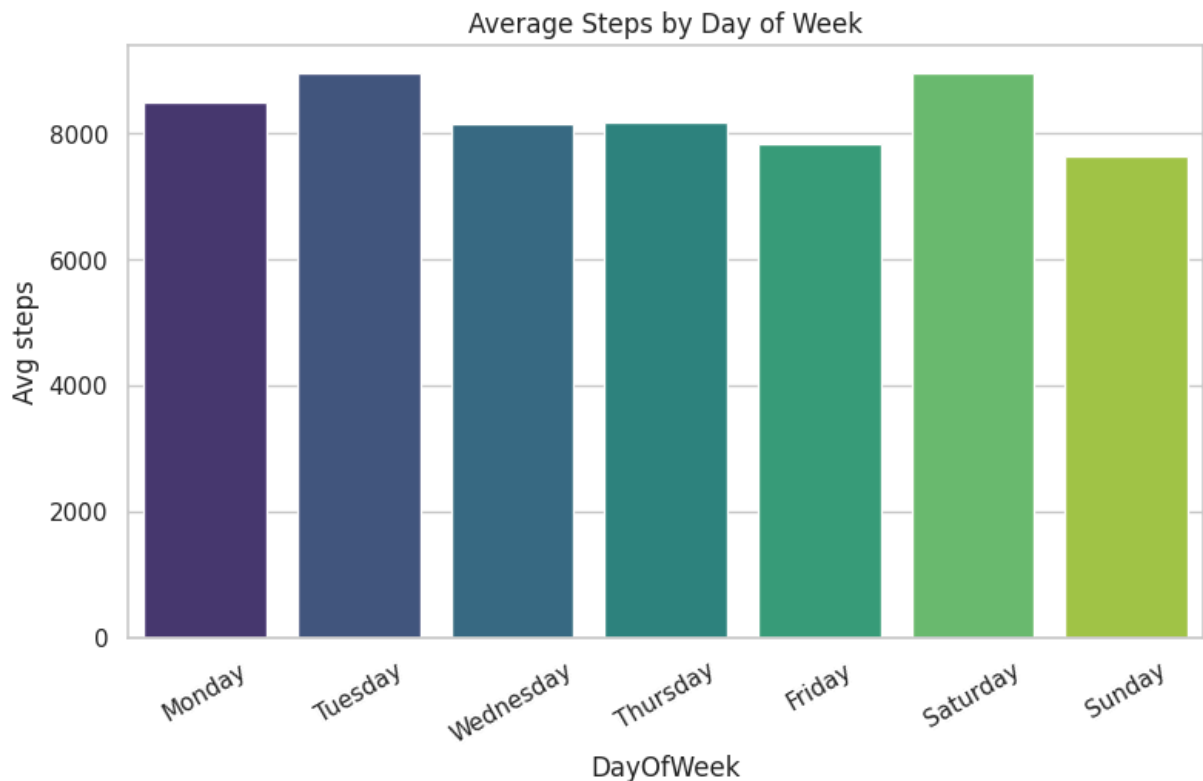


Correlation: 0.562

Insight: steps and calories are **positively correlated (~0.59)** — more steps clearly means more calories burned. This supports gamifying step goals as a route to the health outcomes users care about.

Chart 4 — Activity by Day of Week

```
In [9]: order=['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
dow=activity_worn.groupby('DayOfWeek')['TotalSteps'].mean().reindex(order)
sns.barplot(x=dow.index, y=dow.values, palette='viridis')
plt.xticks(rotation=30); plt.ylabel('Avg steps'); plt.title('Average Steps by Day o
```

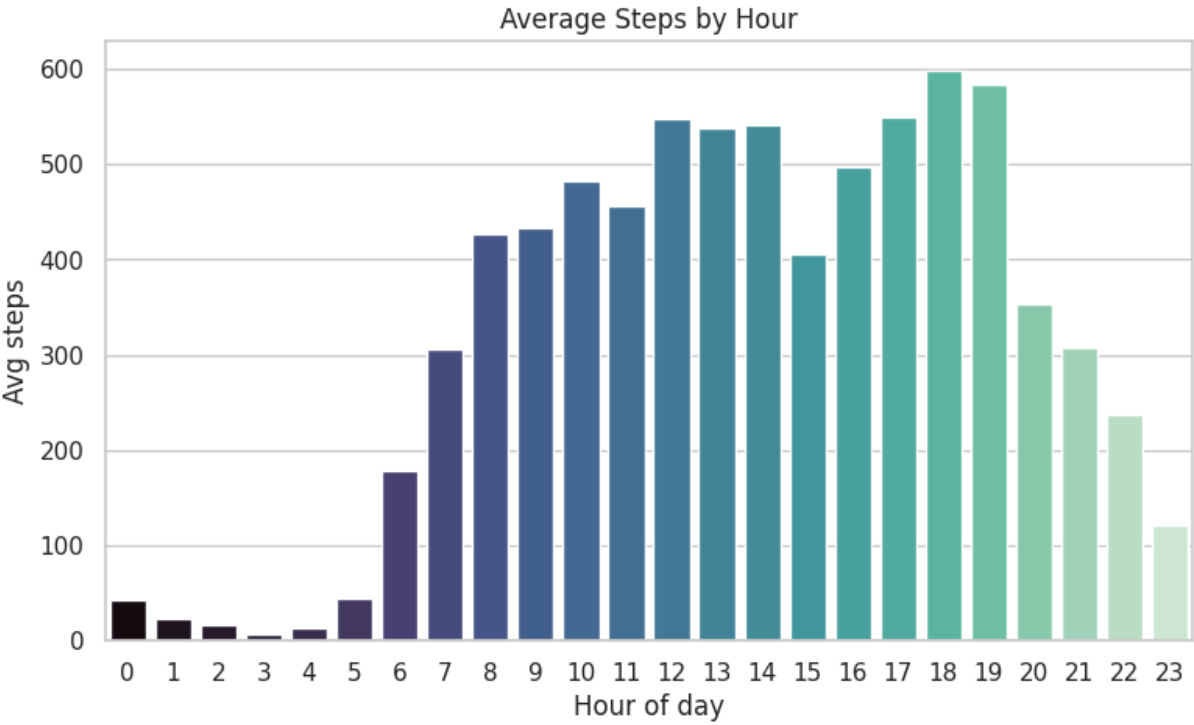


Insight: activity is fairly steady across the week, with small peaks mid-week and on Saturday. Marketing nudges can be scheduled to reinforce the lower-activity days.

4. Hourly Activity Patterns

Chart 5 — Average Steps by Hour of Day

```
In [10]: hourly = hsteps.groupby('Hour')['StepTotal'].mean()
sns.barplot(x=hourly.index, y=hourly.values, palette='mako')
plt.xlabel('Hour of day'); plt.ylabel('Avg steps'); plt.title('Average Steps by Hour')
print("Peak hours:", hourly.sort_values(ascending=False).head(3).round(0).to_dict())
```



Peak hours: {18: 599.0, 19: 583.0, 17: 550.0}

Insight: activity **peaks in the early evening (5–7pm)** and around **noon** — likely commutes, lunch walks, and after-work exercise. These are the ideal windows for push-notifications and reminders.

5. Sleep Analysis

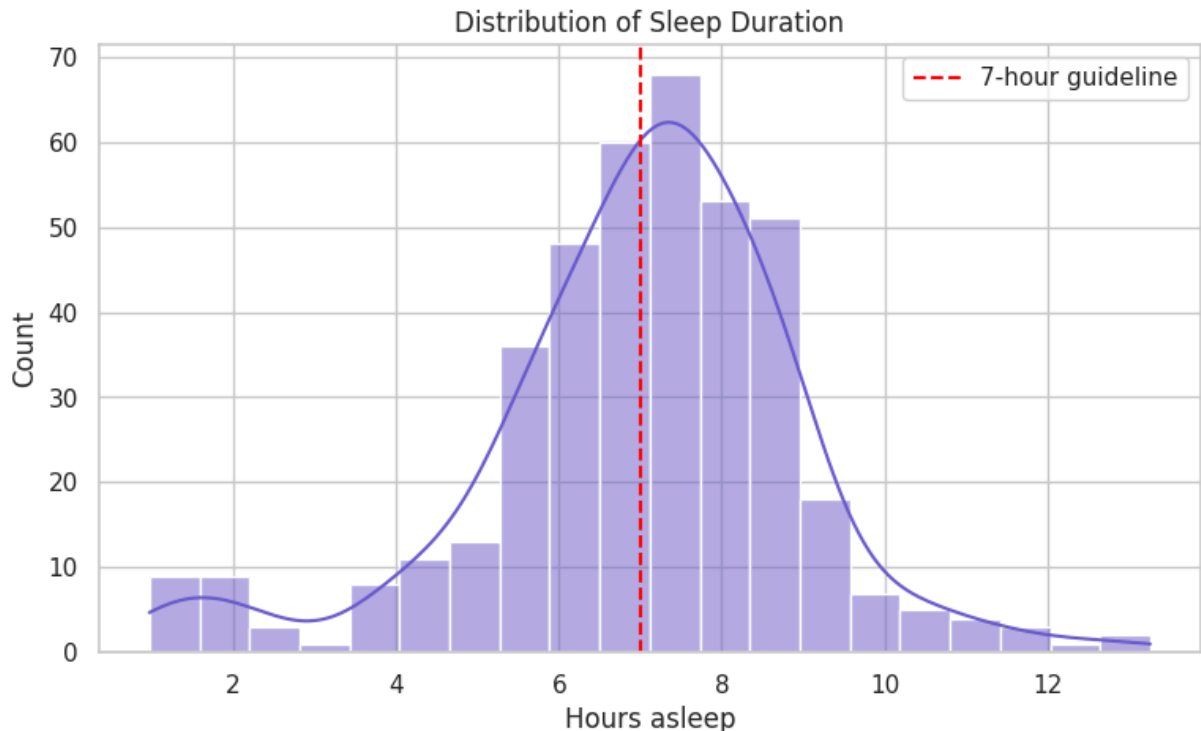
```
In [11]: sleep[['TotalMinutesAsleep', 'TotalTimeInBed', 'SleepEfficiency']].describe().round(1)
```

Out[11]:

	TotalMinutesAsleep	TotalTimeInBed	SleepEfficiency
count	410.0	410.0	410.0
mean	419.2	458.5	91.6
std	118.6	127.5	8.7
min	58.0	61.0	49.8
25%	361.0	403.8	91.2
50%	432.5	463.0	94.3
75%	490.0	526.0	96.1
max	796.0	961.0	100.0

Chart 6 — Sleep Duration Distribution

```
In [12]: sns.histplot(sleep['TotalMinutesAsleep']/60, bins=20, kde=True, color='slateblue')
plt.axvline(7, color='red', ls='--', label='7-hour guideline')
plt.legend(); plt.xlabel('Hours asleep'); plt.title('Distribution of Sleep Duration')
print("Avg hours asleep:", round(sleep['TotalMinutesAsleep'].mean()/60,1))
print("% sleeping < 7 hours:", round((sleep['TotalMinutesAsleep']<420).mean()*100,1))
```



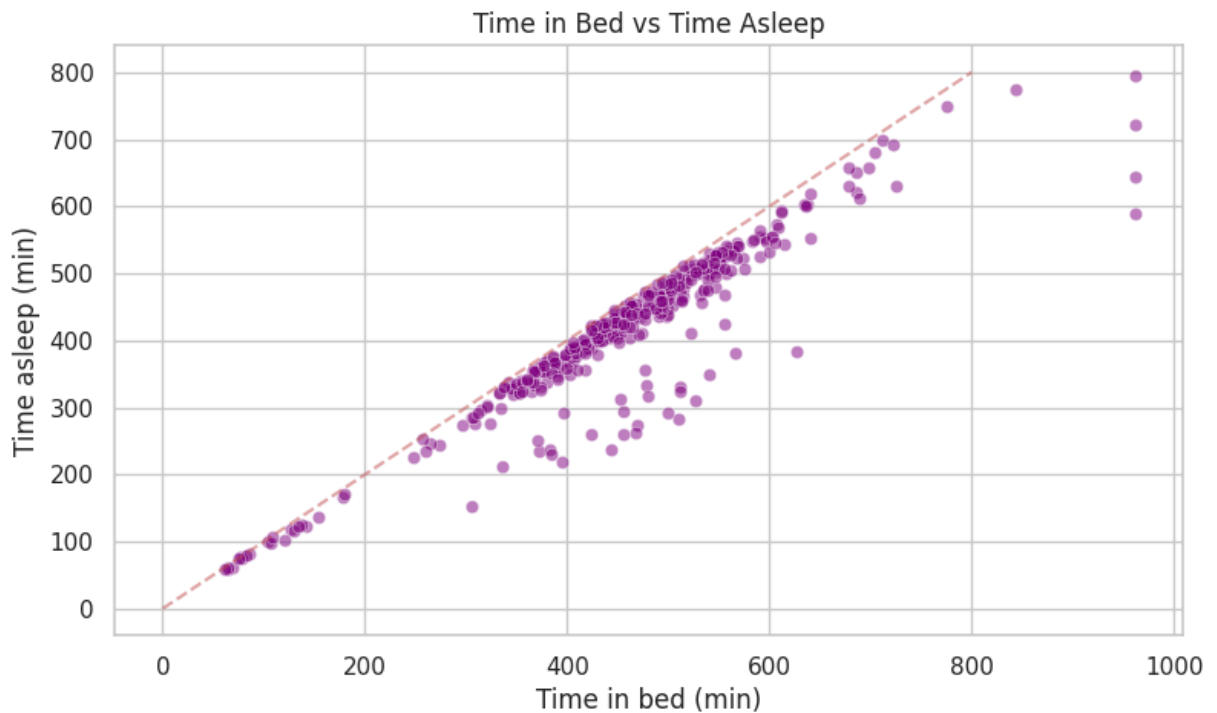
Avg hours asleep: 7.0

% sleeping < 7 hours: 44.1

Insight: users sleep ~**7.0 hours** on average, but ~**44% get less than the recommended 7 hours**. Sleep efficiency is high (~91%), so the issue is *time in bed*, not restlessness — a hook for bedtime-reminder features.

Chart 7 — Time in Bed vs Time Asleep

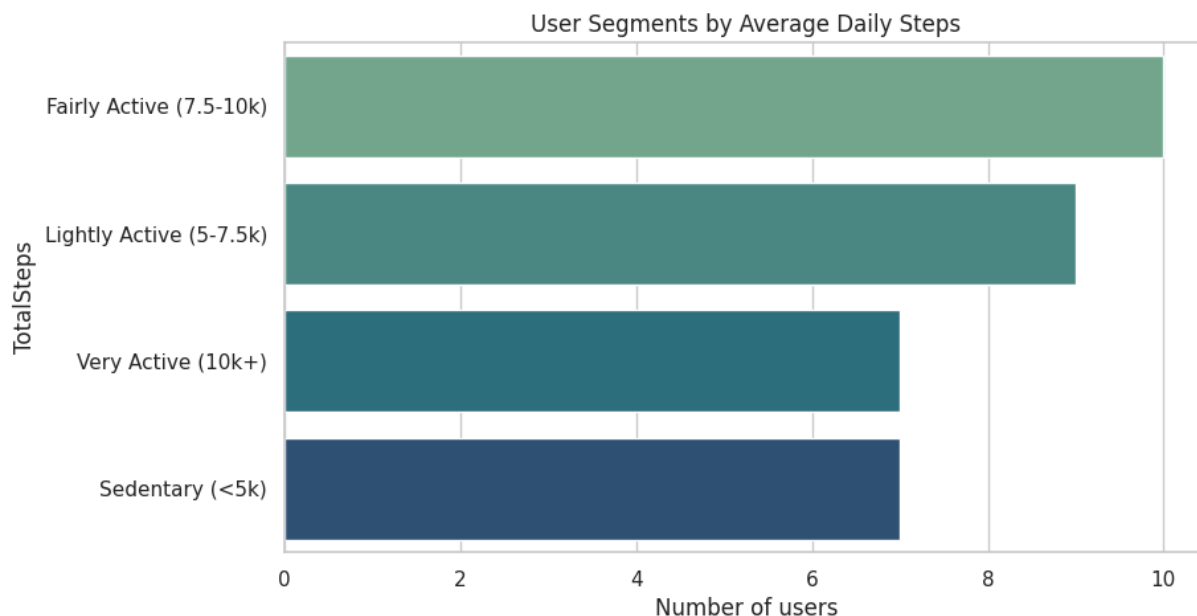
```
In [13]: sns.scatterplot(x='TotalTimeInBed', y='TotalMinutesAsleep', data=sleep, alpha=0.5,
plt.plot([0,800],[0,800], 'r--', alpha=0.5)
plt.xlabel('Time in bed (min)'); plt.ylabel('Time asleep (min)')
plt.title('Time in Bed vs Time Asleep'); plt.show()
```

Insight: points hug the diagonal — once in bed, users fall asleep efficiently. The gap to better sleep is getting to bed earlier, not sleep quality.

6. User Segmentation by Activity Level

```
In [14]: # Segment users by their average daily steps
user_steps = activity_worn.groupby('Id')['TotalSteps'].mean()
def segment(s):
    if s < 5000: return 'Sedentary (<5k)'
    if s < 7500: return 'Lightly Active (5-7.5k)'
    if s < 10000: return 'Fairly Active (7.5-10k)'
    return 'Very Active (10k+)'
seg = user_steps.apply(segment).value_counts()
sns.barplot(x=seg.values, y=seg.index, palette='crest')
plt.xlabel('Number of users'); plt.title('User Segments by Average Daily Steps'); p
seg
```

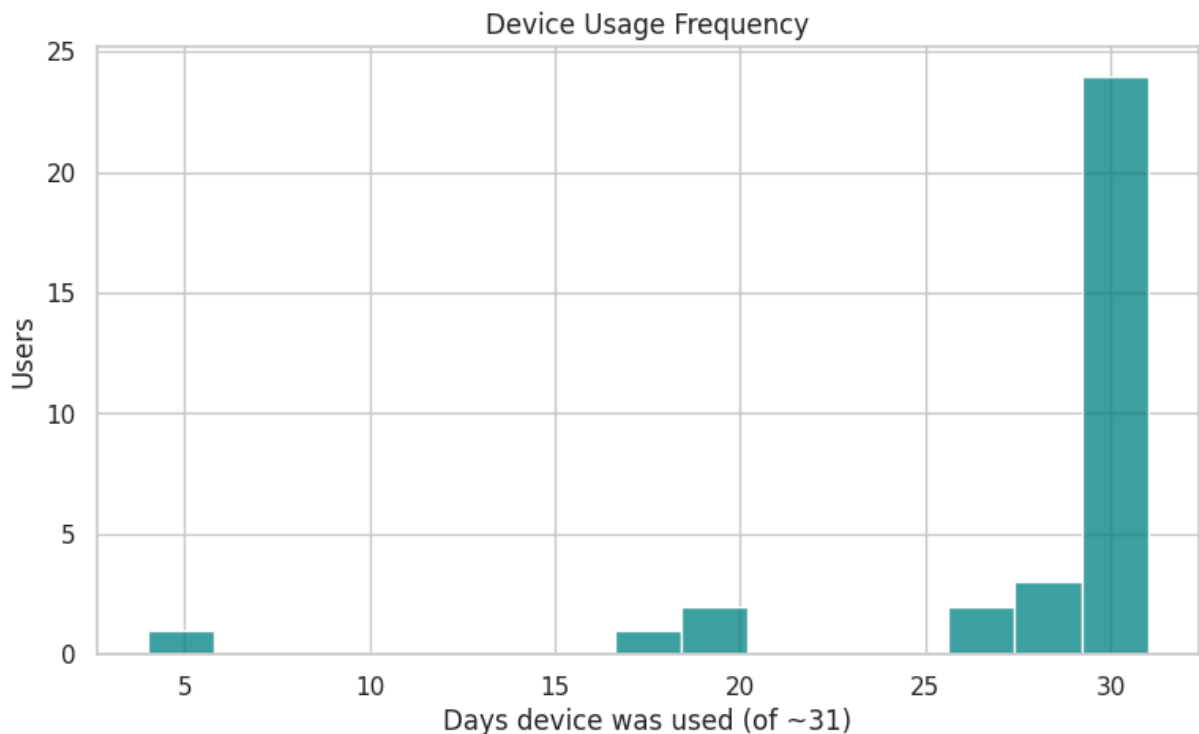


```
Out[14]: TotalSteps
Fairly Active (7.5-10k)    10
Lightly Active (5-7.5k)    9
Very Active (10k+)        7
Sedentary (<5k)           7
Name: count, dtype: int64
```

Insight: users split fairly evenly across activity levels — roughly a quarter each of Sedentary, Lightly, Fairly, and Very Active. Marketing should speak to *all* segments, with the biggest upside in nudging the Sedentary and Lightly Active groups upward.

Chart 8 — Device Usage Frequency

```
In [15]: days_used = activity.groupby('Id')['ActivityDate'].nunique()
sns.histplot(days_used, bins=15, color='teal')
plt.xlabel('Days device was used (of ~31)'); plt.ylabel('Users')
plt.title('Device Usage Frequency'); plt.show()
print("Average days used per user:", round(days_used.mean(),1), "of 31")
```



Average days used per user: 28.5 of 31

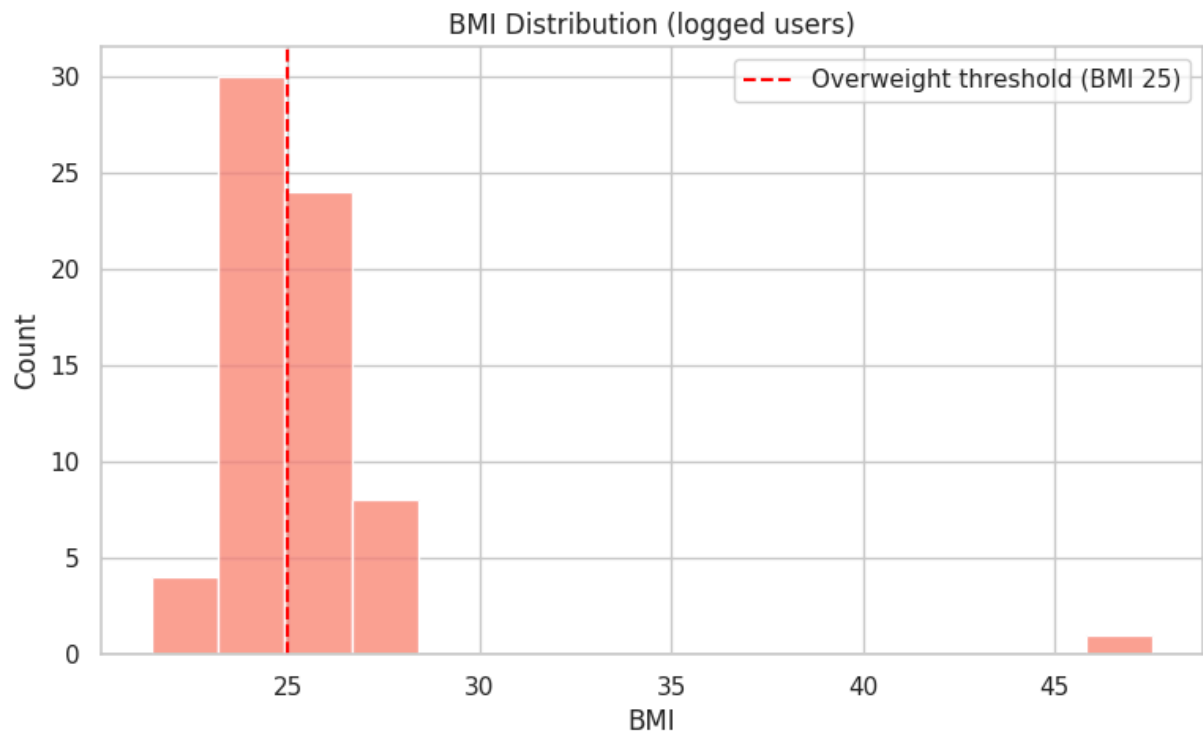
Insight: engagement is **high** — **users wear the device ~28 of 31 days**. People who buy a tracker use it consistently, so retention features (streaks, daily insights) will land well.

Chart 9 — Weight Logging (small sample)

```
In [16]: print("Users who logged weight:", weight['Id'].nunique(), "of 33")
print("Manual vs auto logs:", weight['IsManualReport'].value_counts().to_dict())
sns.histplot(weight['BMI'], bins=15, color='salmon')
plt.axvline(25, color='red', ls='--', label='Overweight threshold (BMI 25)')
plt.legend(); plt.xlabel('BMI'); plt.title('BMI Distribution (logged users)'); plt.
```

Users who logged weight: 8 of 33

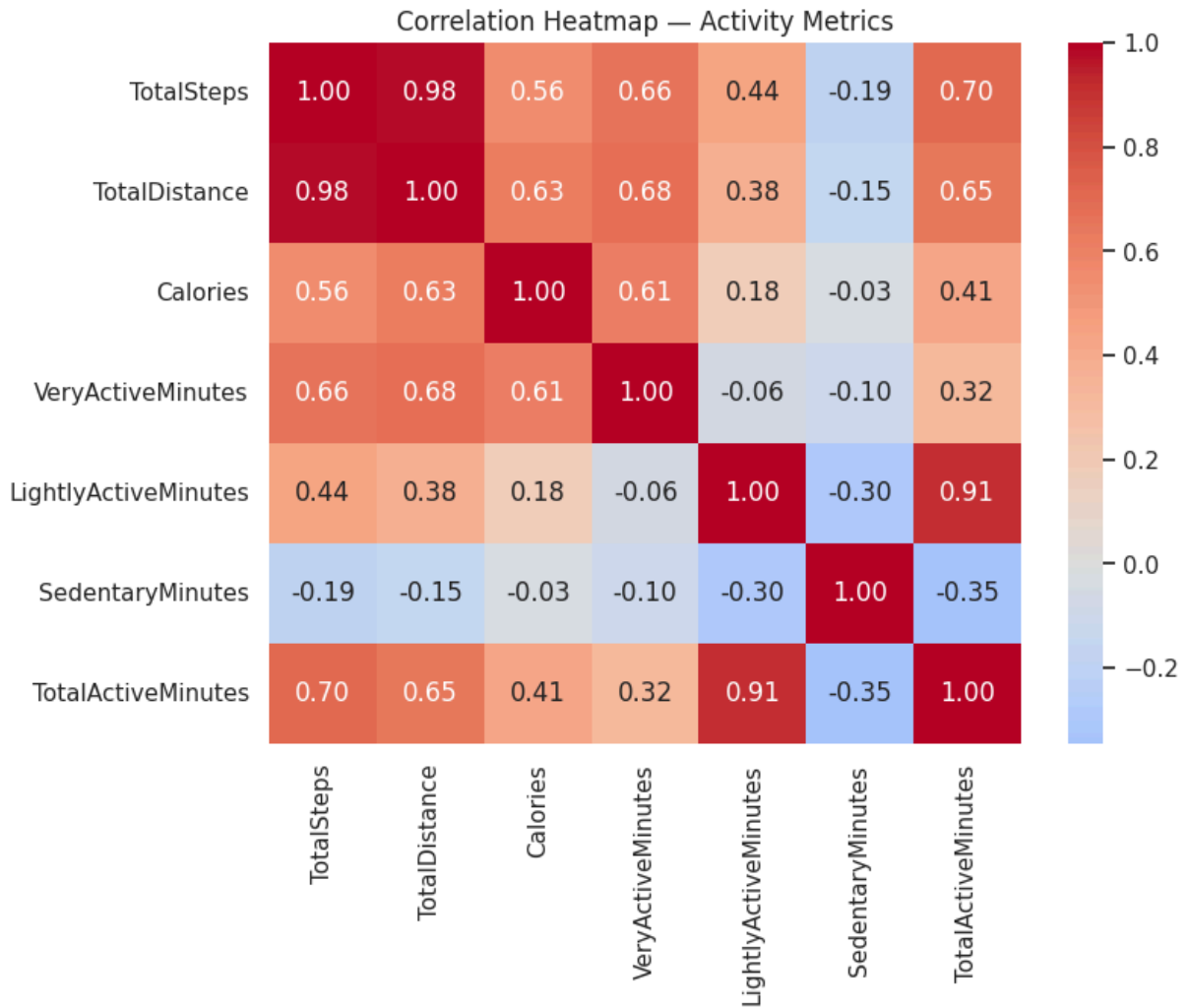
Manual vs auto logs: {True: 41, False: 26}



Insight: only 8 of 33 users logged weight, and most logs were manual. Weight tracking is heavily underused — a clear opportunity for Bellabeat to make weight/BMI logging effortless and engaging.

Chart 10 — Correlation Heatmap

```
In [17]: cols=['TotalSteps','TotalDistance','Calories','VeryActiveMinutes',
               'LightlyActiveMinutes','SedentaryMinutes','TotalActiveMinutes']
plt.figure(figsize=(8,6))
sns.heatmap(activity_worn[cols].corr(), annot=True, fmt='.2f', cmap='coolwarm', cen
plt.title('Correlation Heatmap - Activity Metrics'); plt.show()
```



Insight: steps, distance, and active minutes move together and all relate positively to calories. Sedentary minutes are negatively related to activity — confirming that reducing sedentary time is the lever for healthier outcomes.

7. Conclusions & Bellabeat Marketing Recommendations

What the data tells us about smart-device users:

- They are **highly sedentary** (~16.5 hrs/day) and **fall short of 10,000 steps on ~68% of days**.
- Activity **peaks in the early evening (5–7pm)** and at **noon**.
- They sleep ~7 hours, but ~44% **under-sleep**, and the gap is *getting to bed*, not sleep quality.
- **Engagement is high** (device worn ~28/31 days), but **weight logging is rare** (only 8/33 users).

High-level marketing recommendations for Bellabeat:

1. **Movement reminders & step goals** — most users are sedentary and miss 10k steps. Promote gentle "time to move" nudges and personalised step goals as a headline feature.
2. **Time notifications for the evening peak** — schedule motivational prompts around **noon and 5–7pm**, when users are already active, to reinforce the habit.
3. **Sleep / bedtime coaching** — with ~44% under-sleeping due to late bedtimes, market a **bedtime-reminder and wind-down** feature.
4. **Make weight tracking effortless** — weight logging is heavily underused; automatic or one-tap logging plus encouragement could unlock a new engagement loop.
5. **Lean into high retention** — users wear devices almost daily, so **streaks, daily insights, and rewards** will resonate and deepen loyalty.

Together these turn the observed habits — sedentary days, evening activity peaks, late bedtimes, and strong daily engagement — into concrete, data-backed marketing strategies for Bellabeat's women-focused wellness products.